

# Do Causally Informed Graphs Improve Domain Generalisation in Anomaly Detection?

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**Abstract.** Cyber-physical systems rely on anomaly detection to maintain safe and reliable operation. However, the notion that one model fits all rarely holds in practice, as industrial systems operate across multiple domains with varying system dynamics that can alter the relationships observed in the data. Since the underlying system remains unchanged across domains, causal knowledge may provide a more stable basis for learning transferable representations. Graph-based approaches offer a natural way to model relationships among process variables, but most existing methods capture training-data correlations that are often tied to domain-specific dynamics and may fail to generalise in out-of-domain. In this work, we investigate whether causally informed graphs improve domain generalisation compared to data-driven graph construction using the Causal Chambers dataset and evaluate both approaches across multiple unseen domains. The results show that incorporating causally informed graph structures with graph learning achieves more robust performance across unseen domains and stronger out-of-domain anomaly detection than purely data-driven graph construction. These findings suggest that causal knowledge helps graph-based anomaly detection models capture stable relationships that transfer across domains with different system dynamics.

**Keywords:** Anomaly Detection · Domain Generalisation · Causally Informed Graphs.

## 1 Introduction

Cyber-physical systems (CPS) increasingly monitor and control industrial processes in manufacturing, automotive, and energy systems. These systems connect sensing, computation, actuation, and physical dynamics; therefore, failures are not only computational errors but may also propagate through the physical process and lead to safety risks, equipment damage, production downtime, or cascading failures [14, 10]. As modern CPS generate multivariate sensor streams,

anomaly detection has become a central component of industrial monitoring, aiming to identify deviations from normal behaviour before they develop into faults, attacks, or system-level failures [3, 12, 4].

Anomaly detection in industrial time series is commonly formulated using reconstruction or forecasting errors, where deviations between expected and observed behaviour indicate abnormal operation. Forecasting-based methods are particularly suitable for CPS because anomalies often disrupt the expected temporal evolution of sensor signals. However, many temporal models focus mainly on individual or sequential signal patterns and do not explicitly model relationships between sensors.

Graph-based anomaly detection addresses this limitation by representing sensors as nodes and their relationships as edges, enabling graph neural networks (GNNs) to model dependencies between variables more explicitly. Methods such as MTAD-GAT and Graph Deviation Networks (GDN) demonstrate the benefits of incorporating graph attention and learned sensor relationships for multivariate time-series anomaly detection [19, 5], while graph structure learning (GSL) has also proven effective for multivariate forecasting by capturing latent dependencies among variables [18].

However, the effectiveness of these approaches depends heavily on how the graph is constructed. Existing methods typically rely on fully connected graphs or graph structure learning techniques that infer edges from data. While fully connected graphs may introduce noisy or irrelevant dependencies, graph structure learning infers relationships from statistical dependencies observed in the training-domain data. Under domain shift, these statistical dependencies may change even when the underlying physical process remains the same, causing the learned graph to capture domain-specific associations rather than invariant system mechanisms and reducing anomaly detection reliability [1, 11, 17].

Causal reasoning offers a potential solution. Causal relationships are often more stable across environments than purely statistical associations [13, 16], and invariant prediction studies suggest that models based on causal mechanisms can generalise better under distribution shifts [13, 15]. Since many industrial sensor interactions are governed by physical processes and control logic, causally informed graphs may provide a useful inductive bias for anomaly detection. However, their effectiveness relative to fully connected priors and data-driven structure learning approaches remains largely unexplored.

In this work, we investigate whether causally informed graph structures improve the robustness of graph-based anomaly detection under domain shift. We formulate anomaly detection as a forecasting problem for multivariate time series and keep the model architecture fixed while varying the graph representation and graph construction strategies. Specifically, we compare fully connected and causally informed graph priors under fixed, learned, and hybrid graph refinement strategies, isolating the effects of structural assumptions. Results from the Causal Chambers dataset show that causally informed and hybrid graph structure learning can improve robustness under domain shift compared with dense graph construction.

## 2 Related Work

Industrial anomaly detection models are often trained under one operating condition but deployed under different loads, environmental settings, or control regimes. This creates a domain generalisation problem, where the model must perform well on unseen target domains without using target-domain data during training [17]. Existing domain generalisation methods, such as Risk Extrapolation, encourage models to rely on patterns that remain stable across environments [9]. However, these methods usually focus on the learned features or training objective, rather than on the graph structure used to model relationships between sensors.

Graph-based anomaly detection represents sensors as nodes and inter-sensor dependencies as edges. Methods such as MTAD-GAT and GDN show that modelling these dependencies can improve multivariate time-series anomaly detection [19, 5]. However, the choice of graph construction remains important. Fully connected graphs allow all sensors to exchange information, but may introduce noisy relationships. Learned graphs are more flexible, but their edges reflect the statistical properties from the training data which may not persist when the sensor dynamics change under new operating conditions.

Causal inference provides one way to define such stable relationships. Invariant prediction argues that causal mechanisms are more likely to remain stable across environments than purely statistical correlations [13]. Related work on causal transfer learning and causal representation learning similarly suggests that models based on stable mechanisms can improve generalisation under distribution shift [16, 15]. In multivariate time-series anomaly detection, recent methods such as Entropy Causal Graphs and GCAD also use causal or Granger-causal relationships to improve detection and interpretability [6]. These works show the value of causal structure, but they do not directly test whether causally informed graph priors improve domain generalisation when the forecasting model and anomaly scoring method are kept fixed.

This paper addresses that gap by comparing fully connected and causally prior together with fixed, learned, and hybrid graph refinement strategies within the same forecasting-based anomaly detection framework. By changing only the graph construction, the study isolates whether causal graph priors improve robustness to unseen industrial operating domains.

## 3 Methodology

### 3.1 Problem Formulation

Let a multivariate time series  $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T\}$ , where  $\mathbf{x}_t \in \mathbb{R}^N$  denotes  $N$  sensor variables at time  $t$ . Given a historical window of length  $w$ , the objective is to learn a forecasting function

$$f : \mathbf{X}_{t-w:t-1} \rightarrow \hat{\mathbf{x}}_t,$$

that predicts the expected behaviour under normal conditions. An anomaly score is computed from the deviation between the prediction  $\hat{\mathbf{x}}_t$  and the actual observation  $\mathbf{x}_t$ .

We study this problem under domain generalisation setting. The model is trained on a source operating domain  $D_s$  and evaluated on unseen target domains  $D_t$ , where the data distributions differ, i.e.,  $P_s(\mathbf{X}) \neq P_t(\mathbf{X})$ . The objective is to learn a forecasting model that generalises across domains without access to target-domain data during training.

### 3.2 Graph Representation Strategies

The relationships among sensor variables are represented as a directed graph

$$G = (V, E)$$

where each node  $v_i \in V$  corresponds to a sensor variable and each edge represents a dependency between two variables. The graph is encoded by a binary adjacency matrix

$$A \in \mathbb{R}^{N \times N}$$

Self-loops are included in all graph representations to preserve node-specific information during message passing.

In this work, the graph representation is characterised by two design choices: *graph construction* and *graph refinement* strategies. The former determines the initial graph topology, while the latter determines whether and how the graph is adapted during training.

#### Graph Construction

*Fully Connected Prior (FC)*: The fully connected prior assumes complete connectivity among sensor variables such that

$$A_{ij} = 1 \quad \forall \quad v_i, v_j$$

This results in a dense graph where every sensor variable can directly exchange information with every other variable.

*Causal Prior*: The causal prior is constructed from prior knowledge of the industrial process, where edges correspond to known causal dependencies between variables. The corresponding adjacency matrix is defined as

$$A_{ij} = \begin{cases} 1, & \text{if } (v_j, v_i) \in E_{\text{causal}}, \\ 0, & \text{otherwise,} \end{cases}$$

where  $E_{\text{causal}}$  denotes the set of predefined causal relationships. Compared with the fully connected graph, this representation restricts connectivity to process-informed dependencies.

### Graph Refinement Strategies

*No Graph Structure Learning (No-GSL):* In the No-GSL configuration, graph refinement is not applied and the predefined graph remains unchanged throughout training,

$$E_{\text{final}} = E_{\text{prior}}$$

*Graph Structure Learning (GSL):* Graph structure learning (GSL) adaptively refines the predefined graph using learnable node embeddings. Each node is associated with an embedding vector

$$z_i \in \mathbb{R}^d$$

and the similarity between nodes is computed using cosine similarity,

$$w_{ij} = \frac{z_i^\top z_j}{\|z_i\|_2 \|z_j\|_2}$$

For each node, the *top-k neighbours* with the highest similarity scores are selected to refine the graph connectivity. In this type of configurations, refinement is constrained by the corresponding graph prior, allowing the model to update edge importance while preserving the original topology.

*Hybrid Learning:* This configuration combines a causally informed graph prior with adaptive graph refinement and aims to discover complementary dependencies that are not represented by the predefined causal structure.

To achieve this, graph refinement is applied to a *pruned fully connected graph*, obtained by removing all causal edges from the fully connected graph. The remaining edges constitute the candidate set for graph refinement, from which the top-*k* connections are selected to form an augmentation edge set, denoted by  $E_{\text{augment}}$ . The final graph is then constructed as

$$E_{\text{final}} = E_{\text{causal}} \cup E_{\text{augment}}$$

By preserving the predefined causal graph while augmenting it with complementary learned connections, hybrid learning combines process-informed knowledge with data-driven graph refinement, enabling the model to capture additional dependencies without altering the existing causal structure.

### 3.3 Forecasting-based Anomaly Detection Framework

Figure 1 illustrates the forecasting-based anomaly detection framework used in this study. A sliding window of multivariate sensor observations is first represented as a graph using one of the graph construction strategies described in Section 3.2. Depending on the selected configuration, the graph is either kept

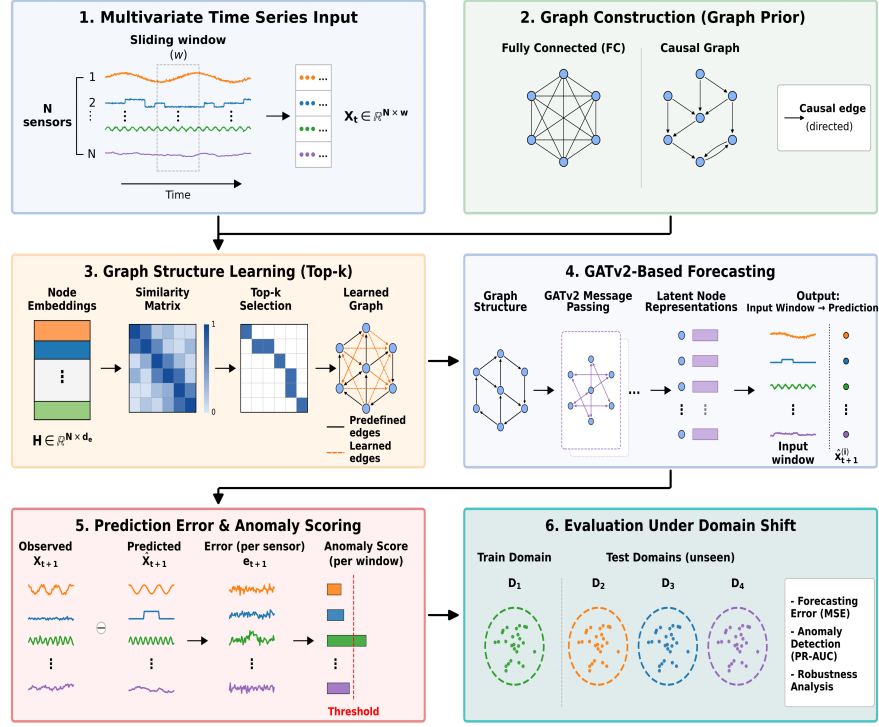


Fig. 1: Overview of the forecasting-based anomaly detection framework used in this study, illustrating graph construction, graph refinement, forecasting, anomaly scoring, and evaluation under domain shift.

fixed, refined through graph structure learning (GSL), or augmented using the hybrid learning strategy.

The resulting graph representation is processed by a graph attention network based on GATv2 [2], which employs a dynamic attention mechanism to adaptively weight neighbouring sensor interactions during message passing. To isolate the effect of graph construction and graph refinement strategies, the GATv2 forecasting architecture is kept fixed across all experimental configurations, ensuring that the observed differences arise from the graph representation rather than the choice of forecasting model. The model is trained by minimizing the mean squared error (MSE) between the predicted and observed sensor values. For configurations employing graph structure learning, the learnable node embeddings are jointly optimized with the forecasting network using the forecasting loss. After each parameter update, cosine similarities between the updated node embeddings are recomputed to refine the graph connectivity, allowing graph refinement and forecasting to be learned end-to-end.

During inference, the model predicts the next multivariate observation from the historical window. The forecasting error for each sensor is then computed as the deviation between the predicted and observed values. Following the Graph Deviation Scoring Framework from GDN [5], these node-wise prediction errors are normalised using statistics estimated from the training data and aggregated into a single anomaly score for each time step, where higher scores indicate a greater likelihood of anomalous behaviour.

## 4 Experiments

The experiments are designed to answer the following research questions:

- **RQ1 (Impact of Graph Construction):** Does replacing a fully connected graph with a causally informed graph improve anomaly detection performance under domain shift?
- **RQ2 (Impact of Graph Refinement):** How do different graph refinement strategies, including graph structure learning and hybrid learning, influence the robustness of causally informed graph representations under domain shift?

### 4.1 Datasets and Domain Setup

Experiments are conducted on the Causal Chambers dataset [7] and the Secure Water Treatment (SWaT) dataset [8].

*Causal Chambers:* Causal Chambers is a real-world physical testbed designed for causal machine learning and out-of-distribution evaluation in cyber-physical systems [7]. In this work, we use the wind tunnel chamber, which provides multivariate sensor measurements together with an expert-defined causal graph describing the dependencies among system variables. We consider the 16 sensor variables that capture the underlying system dynamics, excluding control and auxiliary variables. The testbed enables data generation under configurable operating conditions while preserving the underlying physical system, making it well suited for isolating the effect of graph construction strategies under domain generalisation.

For anomaly detection, anomalous operating conditions are defined based on the hatch position of the wind tunnel. When the hatch is closed, the chamber operates under nominal pressure dynamics, whereas opening the hatch alters the airflow and reduces the chamber pressure, resulting in a different operating regime. The forecasting models are trained exclusively on data collected under nominal operation, while the hatch-based labels are used only for evaluation.

To evaluate domain generalisation, we generate four operating domains ( $D_1$ – $D_4$ ) that differ in their input and output fan loads, signal amplitudes, and operating frequencies, as summarised in Table 1. Models are trained on the nominal source domain  $D_1$  and evaluated on all four domains, where  $D_2$ – $D_4$  represent

Table 1: Qualitative operating conditions and distribution shift of the Causal Chambers domains.

|  | Domain | Input | Output | Amplitude | Frequency | MMD   |
|--|--------|-------|--------|-----------|-----------|-------|
|  | $D_1$  | Step  | Step   | Mixed     | High      | 0.465 |
|  | $D_2$  | Step  | Sine   | Mixed     | High      | 0.624 |
|  | $D_3$  | Sine  | Step   | High      | Low       | 0.745 |
|  | $D_4$  | Sine  | Sine   | Low       | Low       | 0.831 |

previously unseen operating conditions. The table also reports the Maximum Mean Discrepancy (MMD) between  $D_1$  and each target domain, showing a progressive increase in distribution shift while preserving the underlying process structure.

*SWaT*: The Secure Water Treatment (SWaT) dataset is a widely used benchmark for industrial anomaly detection, consisting of multivariate sensor and actuator measurements collected from a scaled water treatment plant under normal and attack scenarios [8]. Unlike Causal Chambers, SWaT contains data from only a single operating condition and therefore does not provide multiple domains for evaluating domain generalisation. Consequently, SWaT is used only for in-distribution evaluation, while cross-domain robustness is evaluated on the Causal Chambers dataset.

## 4.2 Baselines

To evaluate the contribution of graph construction and graph refinement, we consider the following five graph representations:

1. **Fully Connected + No-GSL**, which employs a fully connected graph without graph structure learning.
2. **Fully Connected + GSL**, which learns the graph structure from a fully connected prior.
3. **Causal + No-GSL**, which uses the predefined causal graph without refinement.
4. **Causal + GSL**, which applies graph structure learning to the causal graph.
5. **Hybrid Learning**, which preserves the causal graph while augmenting it with selectively learned edges.

The experimental design isolates the influence of graph representation on cross-domain robustness. In particular, the first two configurations serve as data-driven baselines, while the remaining three evaluate the effect of incorporating causal prior knowledge into graph construction and graph refinement under domain shift.

### 4.3 Evaluation Metrics

Model performance is evaluated using the Precision-Recall Area Under the Curve (PR-AUC), F1-score, Mean Squared Error (MSE), and training time per epoch. The F1-score and PR-AUC assess anomaly detection performance, while MSE evaluates forecasting accuracy. Training time per epoch is used to compare the computational efficiency of different graph representations. For the Causal Chambers dataset, these metrics are reported separately for each target domain to assess robustness. All experiments are repeated over 5 independent runs, and the reported results correspond to the average performance.

### 4.4 Experimental Setup

Experiments are conducted on a GPU-enabled system equipped with an NVIDIA Tesla T4 GPU (15 GB VRAM), a 4-core CPU, and 15 GB of system memory. The models are implemented in Python 3.10.12 using PyTorch 2.5.1 and PyTorch Geometric 2.7.0 with CUDA 12.1 for GPU acceleration. Model parameters are optimised using the Adam optimiser with a learning rate of  $1 \times 10^{-2}$  and a batch size of 64, and trained for a maximum of 100 epochs with early stopping based on the validation loss. The forecasting model employs a sliding window of length  $w = 10$  with a stride of 5, a hidden dimension of 16, two GNN layers, a single attention head, and a dropout rate of 0.05. For graph structure learning, the node embedding dimension is set to 16 and the graph sparsity parameter  $k$  is evaluated for value  $\{2\}$ , while the No-TopK configuration retains all candidate neighbours. The forecasting head consists of a single MLP layer.

## 5 Results

### 5.1 Sanity check

Our forecasting model is a variation the Graph Deviation Network (GDN) architecture [5], where the original attention mechanism is replaced by GATv2 attention [2]. To sanity check that this modification does not influence anomaly detection performance, we compare the original GDN implementation with the GATv2-based implementation on the SWaT dataset using identical hyperparameters. As shown in Table 2, both implementations achieve nearly identical performance, justifying the use of GATv2 as the forecasting backbone for all subsequent experiments.

### 5.2 RQ1: Impact of Graph Construction

To investigate the impact of graph construction, we compare the **Fully Connected (FC) + No-GSL** and **Causal + No-GSL** configurations, where both models employ the same forecasting architecture without graph refinement.

As shown in Table 3 and Figure 2a, both graph representations achieve comparable performance on the source domain ( $D_1$ ). Under domain shift, the fully

Table 2: Comparison between the original GDN attention mechanism and the GATv2-based implementation on the SWaT dataset (results averaged across 10 runs)

| Model | Precision | Recall | F1-score |
|-------|-----------|--------|----------|
| GDN   | 0.969     | 0.637  | 0.766    |
| GATv2 | 0.959     | 0.639  | 0.767    |

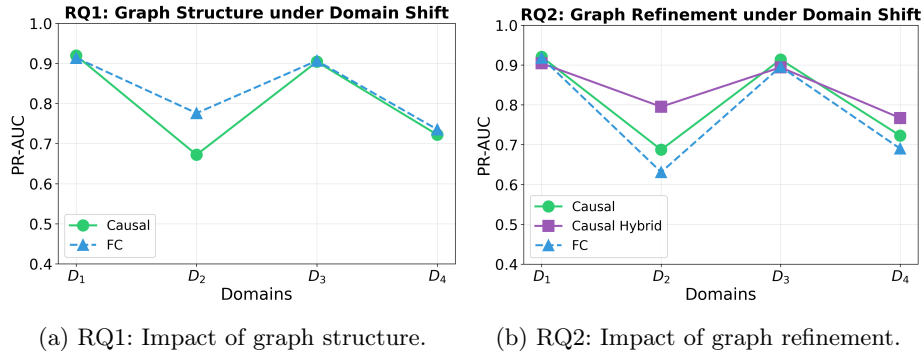


Fig. 2: PR-AUC across the evaluation domains for the graph structure (RQ1) and graph refinement (RQ2) experiments.

connected graph achieves higher PR-AUC and F1-score on  $D_2$ , while the two approaches exhibit nearly identical performance on  $D_3$ . On  $D_4$ , both configurations again produce comparable anomaly detection and forecasting performance, with the causal graph attaining a slightly higher F1-score. The causal graph’s training time per epoch is approximately 1.49x lower than that of the fully connected graph (2.79s vs. 4.16s), reflecting reduced message-passing overhead from its sparser connectivity. Overall, these results suggest that process-informed graph structures can maintain competitive anomaly detection performance while providing improved computational efficiency.

### 5.3 RQ2: Impact of Graph Refinement

To evaluate the effect of graph refinement, we compare the **Fully Connected (FC) + GSL**, **Causal + GSL**, and the **Hybrid Learning** configuration using a graph sparsity parameter of  $k = 2$ .

Table 4 and Figure 2b demonstrate the impact of graph refinement under domain shift. While all three configurations achieve comparable performance on the source domain ( $D_1$ ), the Hybrid Learning strategy achieves the highest PR-AUC on the unseen target domains  $D_2$ , and  $D_4$ , being competitive on domain  $D_3$ . This indicates improved generalisation through the augmentation of the causal graph with selectively learned dependencies. All methods exhibit similar

Table 3: Comparison of the **Fully Connected (FC)** + **No-GSL** and **Causal** + **No-GSL** configurations across the Causal Chambers domains (results averaged across 5 runs per domain).

| Domain | Model           | PR-AUC             | F1-score    | MSE         |
|--------|-----------------|--------------------|-------------|-------------|
| $D_1$  | Causal + No-GSL | <b>0.919±0.012</b> | 0.866±0.005 | 36.356±0.25 |
|        | FC + No-GSL     | 0.913±0.003        | 0.868±0.003 | 36.688±0.24 |
| $D_2$  | Causal + No-GSL | 0.672±0.021        | 0.679±0.024 | 28.05±0.19  |
|        | FC + No-GSL     | <b>0.776±0.106</b> | 0.746±0.020 | 28.30±0.18  |
| $D_3$  | Causal + No-GSL | 0.904±0.011        | 0.817±0.002 | 44.44±0.31  |
|        | FC + No-GSL     | <b>0.907±0.006</b> | 0.818±0.002 | 44.85±0.29  |
| $D_4$  | Causal + No-GSL | 0.722±0.016        | 0.818±0.009 | 21.22±0.15  |
|        | FC + No-GSL     | <b>0.735±0.032</b> | 0.804±0.004 | 21.41±0.14  |

forecasting accuracy, with comparable MSE values. The Hybrid configuration’s training time per epoch (2.75s) is comparable to Causal + GSL (2.88s) and approximately 1.52x lower than Fully Connected + GSL (4.17s), confirming that sparse causal priors reduce computational overhead even when augmented with learned edges. These results suggest that combining causal prior knowledge with selective data-driven graph refinement provides an effective balance between robustness and computational efficiency.

#### 5.4 Ablation Study: Effect of Graph Sparsity

To investigate the influence of graph sparsity on anomaly detection performance, we evaluate the graph refinement strategies using different values of the graph sparsity parameter  $k$ , which determines the number of neighbours retained during graph refinement. Smaller values of  $k$  produce sparser graphs, while larger values allow denser connectivity and increased message passing.

Figures 3 illustrate the effect of the graph sparsity parameter  $k$  on anomaly detection performance. No single value of  $k$  consistently achieves the highest PR-AUC across all domains, indicating that the optimal graph sparsity is domain-dependent. Nevertheless, relatively sparse configurations ( $k = 2$  or  $k = 3$ ) generally provide competitive performance compared with denser alternatives, with the Hybrid configuration achieving the highest PR-AUC on several unseen target domains. Notably, training time per epoch remains approximately constant across all  $k$  values within each configuration (2.74–2.79s for Hybrid, 2.67–2.98s for Causal + GSL, and 4.17–4.26s for FC + GSL), indicating that varying graph sparsity has a negligible impact on computational efficiency while allowing sparse graphs to achieve competitive anomaly detection performance.

Table 4: Comparison of graph refinement strategies ( $k = 2$ ) across the Causal Chambers domains (results averaged across 5 runs per domain).

| Domain | Model        | PR-AUC             | F1-score    | MSE          |
|--------|--------------|--------------------|-------------|--------------|
| $D_1$  | Causal + GSL | <b>0.920±0.016</b> | 0.856±0.024 | 36.549±0.362 |
|        | FC + GSL     | 0.918±0.007        | 0.864±0.007 | 36.815±0.156 |
|        | Hybrid       | 0.904±0.008        | 0.862±0.021 | 36.413±0.234 |
| $D_2$  | Causal + GSL | 0.687±0.073        | 0.711±0.033 | 28.203±0.280 |
|        | FC + GSL     | 0.631±0.067        | 0.699±0.029 | 28.400±0.121 |
|        | Hybrid       | <b>0.795±0.082</b> | 0.743±0.024 | 28.088±0.182 |
| $D_3$  | Causal + GSL | <b>0.914±0.026</b> | 0.832±0.036 | 44.682±0.445 |
|        | FC + GSL     | 0.896±0.003        | 0.819±0.000 | 45.007±0.193 |
|        | Hybrid       | 0.894±0.006        | 0.819±0.000 | 44.510±0.286 |
| $D_4$  | Causal + GSL | 0.723±0.025        | 0.829±0.013 | 21.338±0.210 |
|        | FC + GSL     | 0.690±0.022        | 0.819±0.014 | 21.513±0.089 |
|        | Hybrid       | <b>0.767±0.086</b> | 0.820±0.013 | 21.250±0.140 |

## 6 Discussion

The primary objective of this study was to investigate whether incorporating causal prior knowledge into graph representations improves the domain generalisation of graph-based anomaly detection. Overall, the results suggest that causal knowledge provides a useful structural prior for graph construction, although its effectiveness depends on how it is incorporated into the refinement process. The following discussion interprets these findings in the context of the two research questions and the ablation study.

The findings of **RQ1** indicate that using causal priors alone does not consistently improve anomaly detection performance over a fully connected graph representation. Compared with **Fully Connected + No-GSL**, the **Causal + No-GSL** configuration achieves comparable and, in some cases, lower PR-AUC, most notably on domain  $D_2$ . A possible explanation is that the predefined causal graph captures the physical cause-and-effect relationships of the system but may omit predictive dependencies exploited by the fully connected graph. Despite this, its sparser connectivity reduces message-passing operations and attention computations, resulting in consistently lower training time. Overall, these findings suggest that process-informed graph construction provides improved computational efficiency while maintaining competitive anomaly detection performance.

The findings of **RQ2** build upon the observations from **RQ1** by demonstrating the benefit of combining causal prior knowledge with data-driven graph refinement. While the **Fully Connected + GSL** and **Causal + GSL** configurations achieve broadly comparable performance across the evaluation domains, the **Hybrid Learning** strategy consistently attains the highest or comparably highest PR-AUC on the unseen target domains. This suggests that causal priors

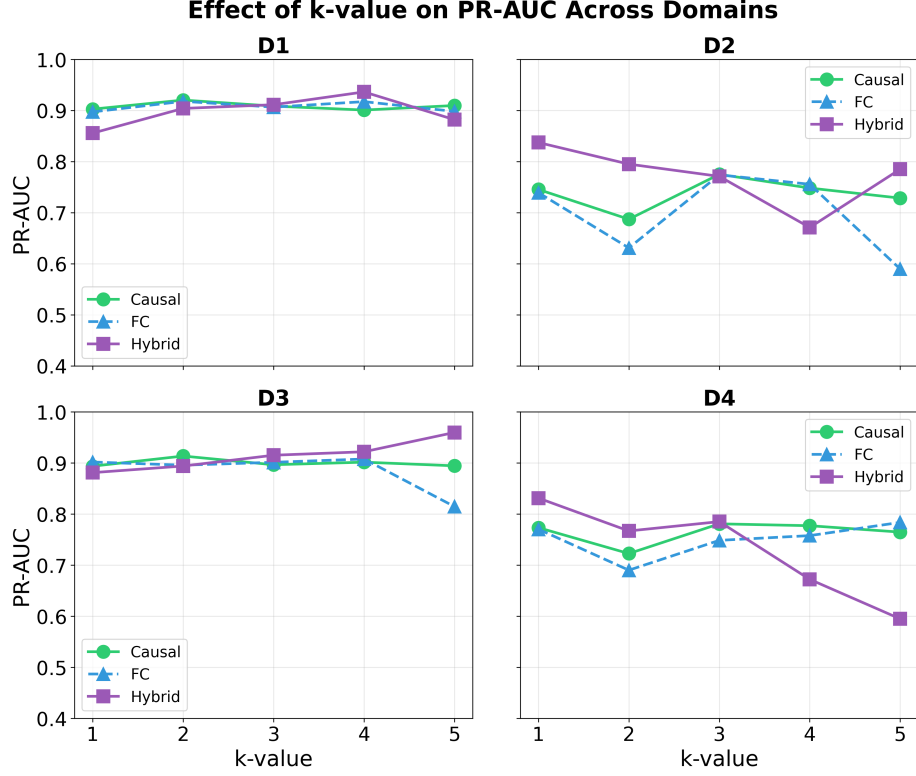


Fig. 3: Ablation study on the effect of graph sparsity parameter  $k$  on anomaly detection performance (PR-AUC)

and graph refinement are complementary, with the former preserving process-informed causal relationships and the latter incorporating additional predictive dependencies that are not encoded in the prior, thereby improving robustness under domain shift. Moreover, the Hybrid configuration retains training times comparable to **Causal** + **GSL**, indicating that augmenting the causal graph with a limited number of learned edges preserves much of the computational efficiency afforded by sparse graph representations. Overall, these findings suggest that combining causal priors with selective graph refinement provides an effective balance between robustness under domain shift and computational efficiency.

The **ablation study** demonstrates that the influence of the graph sparsity parameter is domain dependent, with no single value of  $k$  consistently yielding the best anomaly detection performance across all operating domains. This suggests that the optimal neighbourhood size depends on the characteristics of the target domain, as changes in operating conditions alter the relevance of interactions among sensor variables. Compared with **Fully Connected** + **GSL**, both **Causal** + **GSL** and **Hybrid Learning** consistently require less training time

because their sparse graph representations reduce the number of message-passing operations and attention computations. The Hybrid configuration preserves this efficiency by introducing only a limited number of learned edges on top of the causal graph. Overall, these findings suggest that relatively sparse graph representations provide a practical balance between robustness under domain shift and computational efficiency without extensive parameter tuning.

Another important observation is that the severity of domain shift, as measured by Maximum Mean Discrepancy (MMD), does not directly correspond to anomaly detection performance. Although  $D_3$  and  $D_4$  exhibit larger MMD values than  $D_2$ , all graph representations remain competitive on these domains, suggesting that domain generalisation depends not only on the magnitude of the distribution shift but also on the underlying system dynamics. Consequently, evaluating robustness under domain shift requires considering both statistical distribution metrics and system dynamics.

Despite these promising findings, several limitations should be acknowledged. First, the predefined causal graph may not capture all relevant physical interactions, and although graph refinement can compensate for some missing relationships, its effectiveness remains dependent on the quality of the underlying causal prior. Second, the analysis considers only domain shifts arising from changes in operating conditions under a fixed system structure and may not generalise to scenarios involving structural changes or evolving system topologies.

## 7 Conclusion

This paper investigated whether causally informed graph structures improve domain generalisation in graph-based anomaly detection. To isolate the effect of graph construction strategies, we compared fully connected and causally informed graph priors under fixed, learned, and hybrid graph refinement settings using a forecasting-based framework with a fixed GATv2 architecture on the Causal Chambers dataset.

The results show that causally informed graph representations achieve anomaly detection performance comparable to purely data-driven graph representations while consistently requiring lower training time per epoch. Although predefined causal priors alone do not always improve domain generalisation, combining them with selective graph refinement enables the model to capture additional predictive dependencies while preserving the computational efficiency afforded by sparse graph representations, thereby improving robustness under changing operating conditions. Furthermore, the results indicate that domain shift cannot be fully characterised by quantitative distribution metrics such as Maximum Mean Discrepancy (MMD) alone, as the underlying system dynamics and interactions among sensor variables also influence model transferability. Overall, these findings demonstrate that integrating process-informed causal knowledge with selective graph refinement provides an effective balance between robustness and computational efficiency for domain generalisation in industrial cyber-physical systems.

Future work will extend this study to additional industrial benchmarks, explore the use of causal discovery methods to identify direct relationships and construct more complete causally informed graph representations, and investigate more diverse forms of domain shift involving changes in both operating conditions and system structure. These directions will further improve the applicability, robustness, and generalisability of causally informed graph-based anomaly detection in real-world industrial cyber-physical systems.

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